



C-23-AIM-AI-CM-CAI-CCB-CCN-104

23016

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

FIRST YEAR (COMMON) EXAMINATION

ENGINEERING CHEMISTRY AND ENVIRONMENTAL STUDIES

Time : 3 hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :**
- (1) Answer **all** questions.
 - (2) Each question carries **three** marks.
 - (3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

- 1. What is electronic configuration of chromium and copper?
- 2. Draw the shapes of d-orbitals.
- 3. Define mole. Find the numbers moles in 90 gram of water.
- 4. Define pH calculate the pH of 0.01M HCl solution.
- 5. List out any three applications of electrolysis.
- 6. Explain the sacrificial anode process.
- 7. Write the disadvantages of using hard water in industries.
- 8. Write the composition and application of stainless steel.
- 9. Classify the fuels based on their occurrence.
- 10. Define the terms (a) pollution (b) pollutant and (c) sink.

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[Contd...

PART—B

10×5=50

- Instructions :**
- (1) Answer *any five* questions.
 - (2) Each question carries **ten** marks.
 - (3) Answers should be comprehensive and the criteria for valuation is the content but not the length of the answer.

- 11.** State the postulates of Bohr's atomic theory and its limitations. 10
- 12.** (a) Define equivalent weight acid and calculate the equivalent weight of HCl, H₂SO₄, H₃PO₄. 6
- (b) Classify solutions based on solubility. 4
- 13.** (a) Explain (a) Aufbau principle (b) Pauli's Exclusion principle. 6
- (b) Define buffer solution. Write any three applications of buffer solution. 6
- 14.** Define galvanic cell. Explain its construction and working of galvanic cell. 10
- 15.** Define rusting of iron and explain the mechanism of rusting of iron with chemical equations. 10
- 16.** (a) List out the salts that causing hardness of water (with names and formula. 6
- (b) Explain reverse osmosis process. 4
- 17.** Define elastomer. Explain a method of preparation and uses of (a) Buna-s (b) Neoprene. 10

- 18.** (a) Define the terms (a) Dissolved Oxygen (b) BOD and (c) COD 5
(d) Sink (e) pollutant.
(b) Write a short note on Ozone layer depletion. 5

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